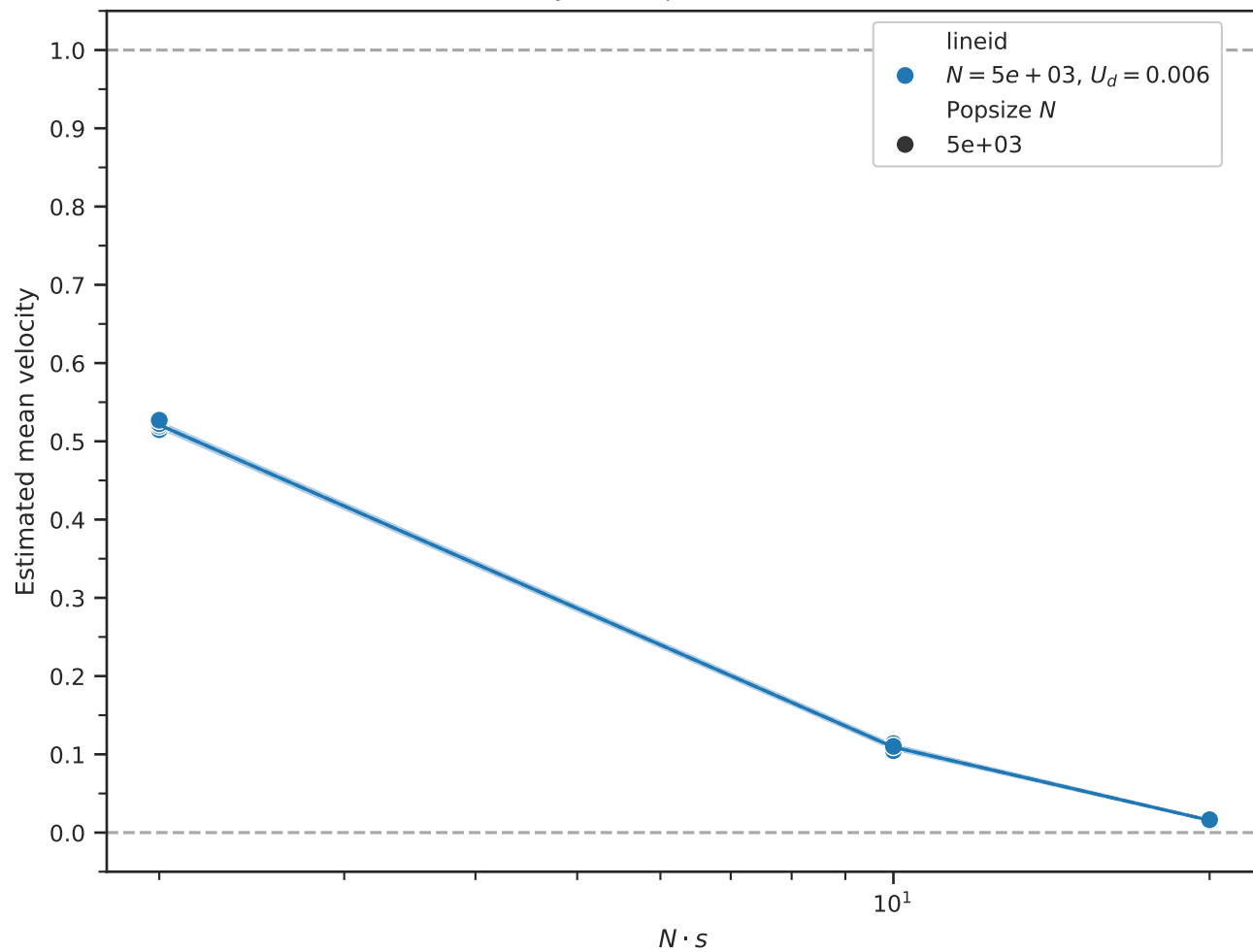
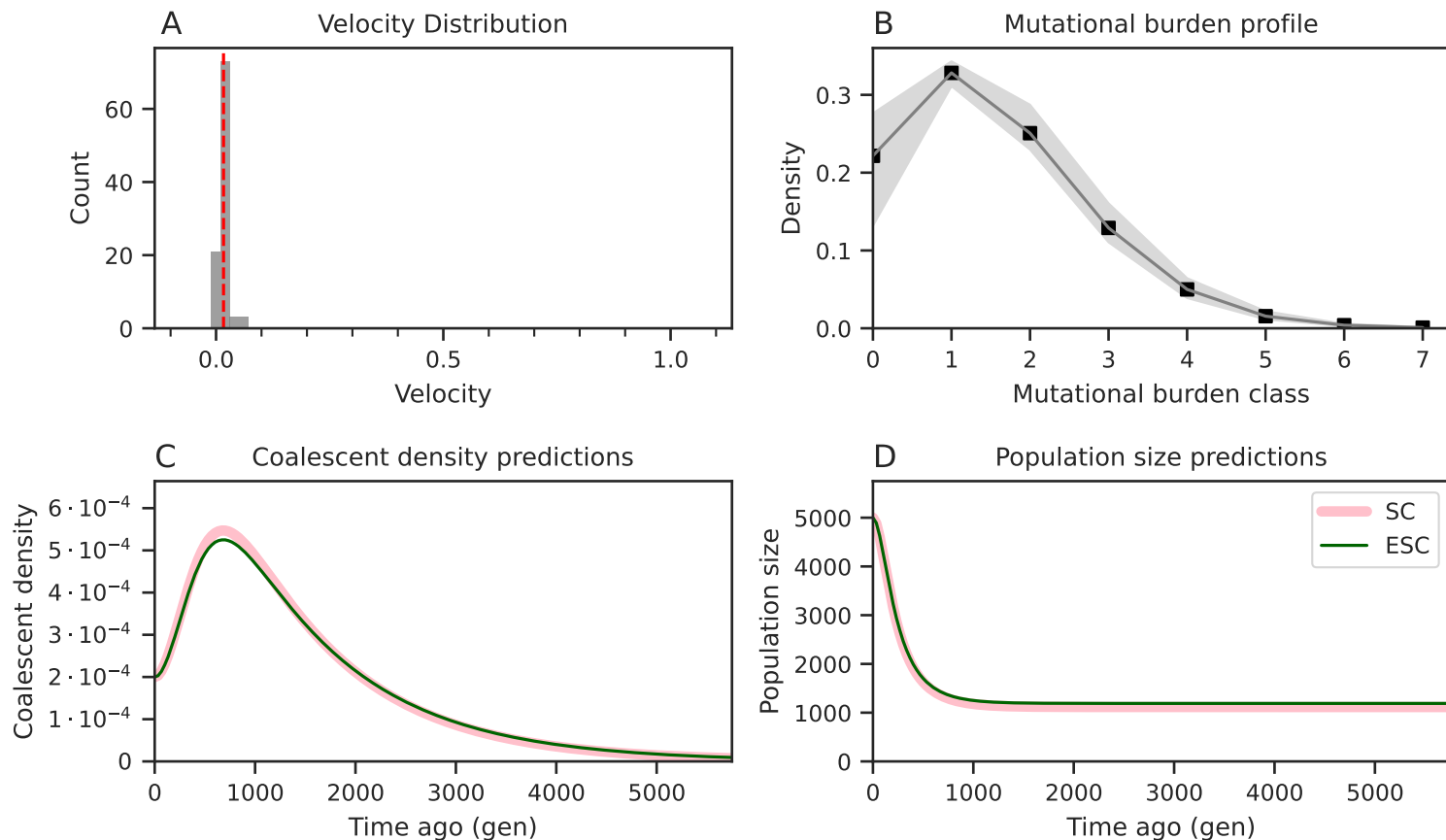


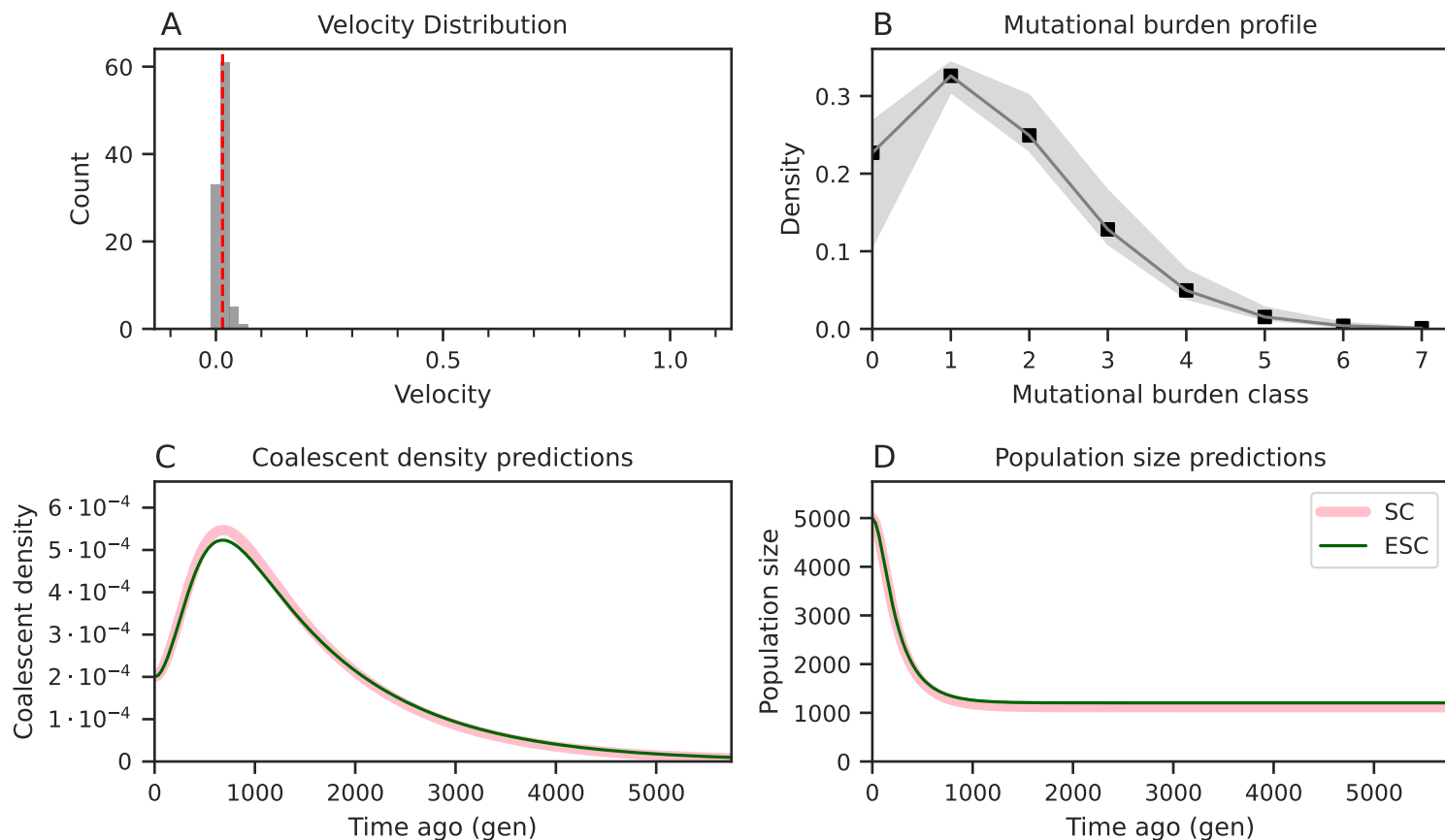
Mean velocity across parameter combinations



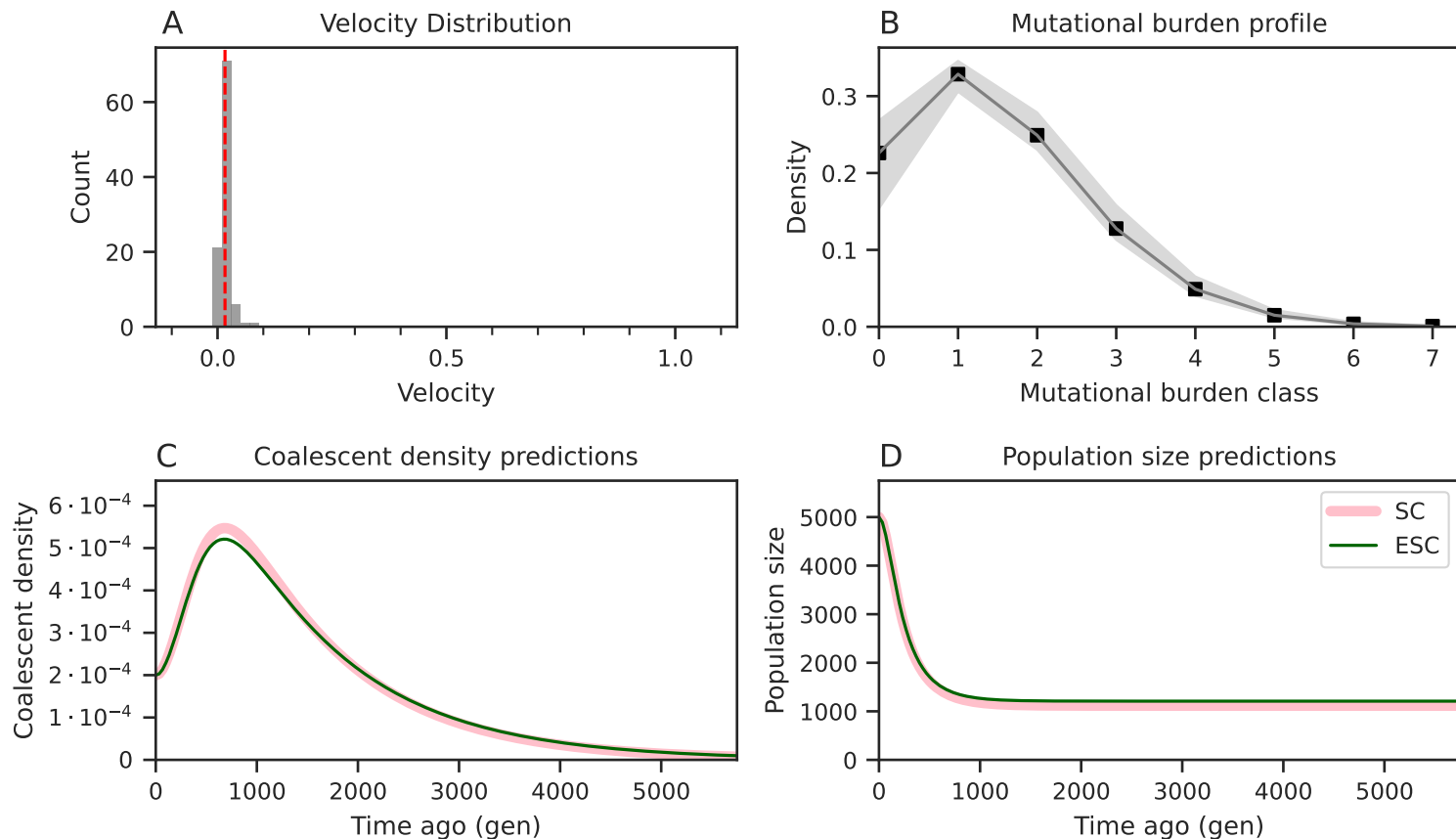
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.002$, $\phi = 4.46$
Mean Velocity= 0.0164 ± 0.0106 , $n = 100$ simulations



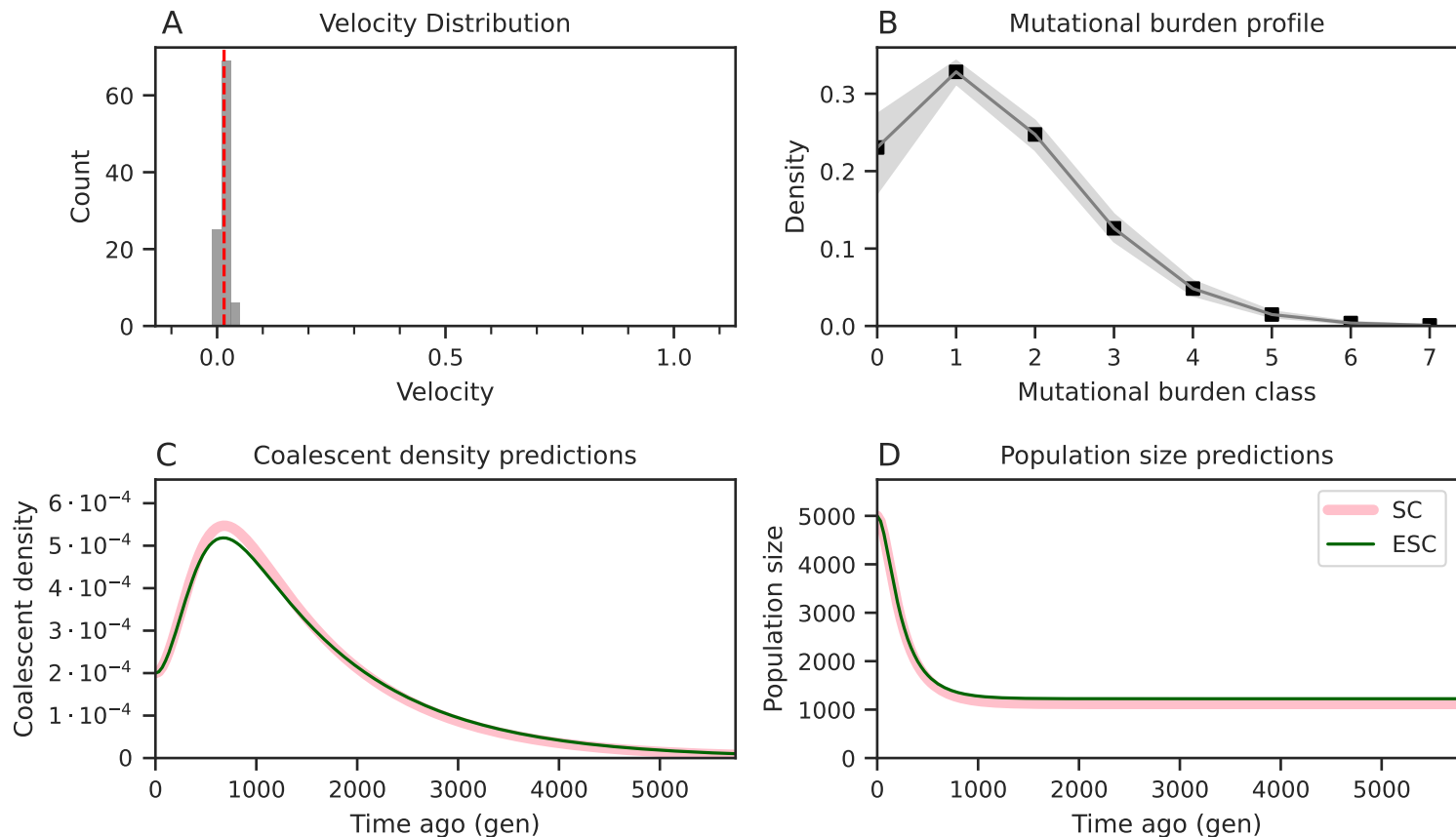
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 4.46$
Mean Velocity= 0.0147 ± 0.0098 , $n = 100$ simulations



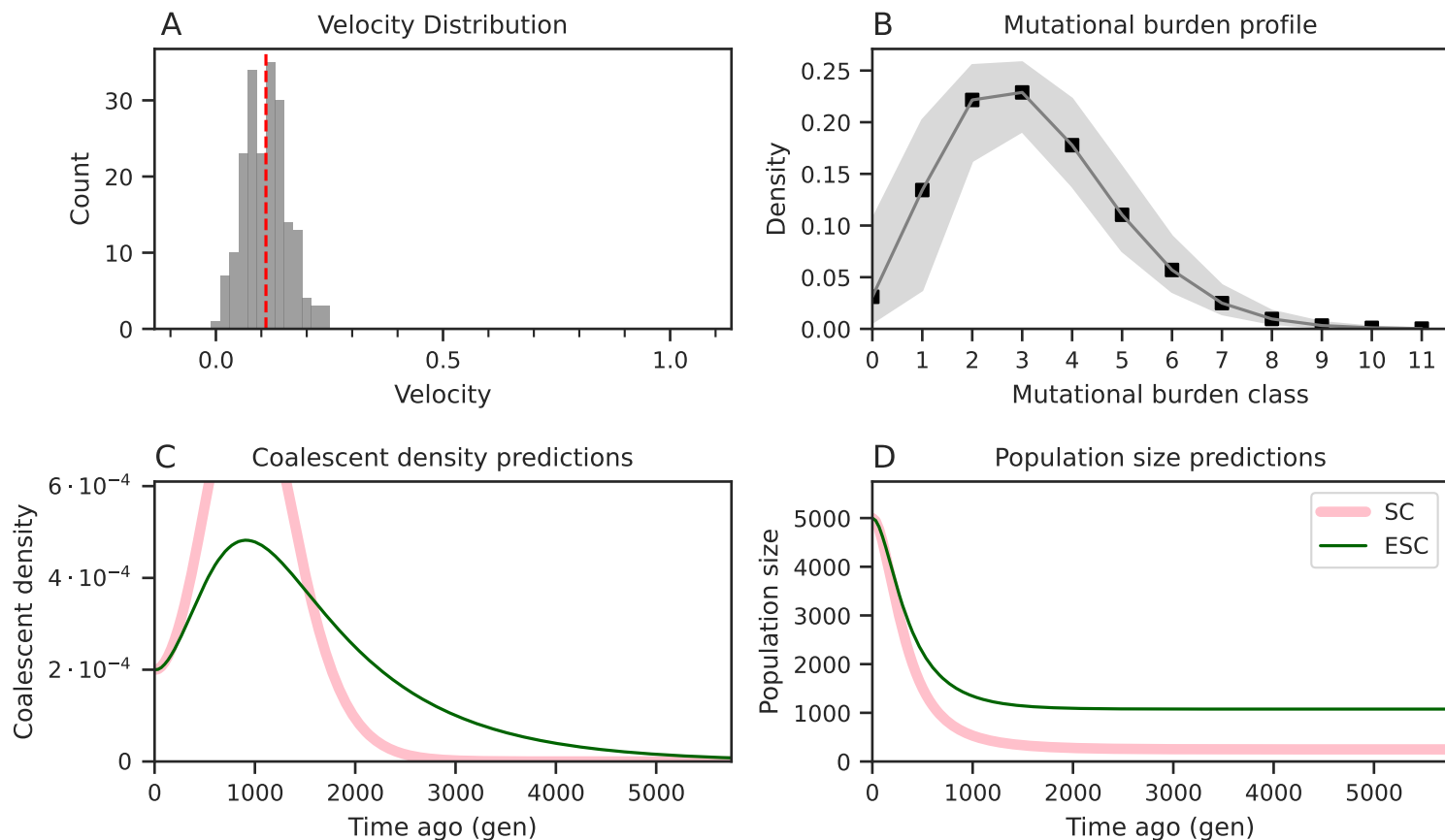
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity= 0.0163 ± 0.0114 , $n = 100$ simulations



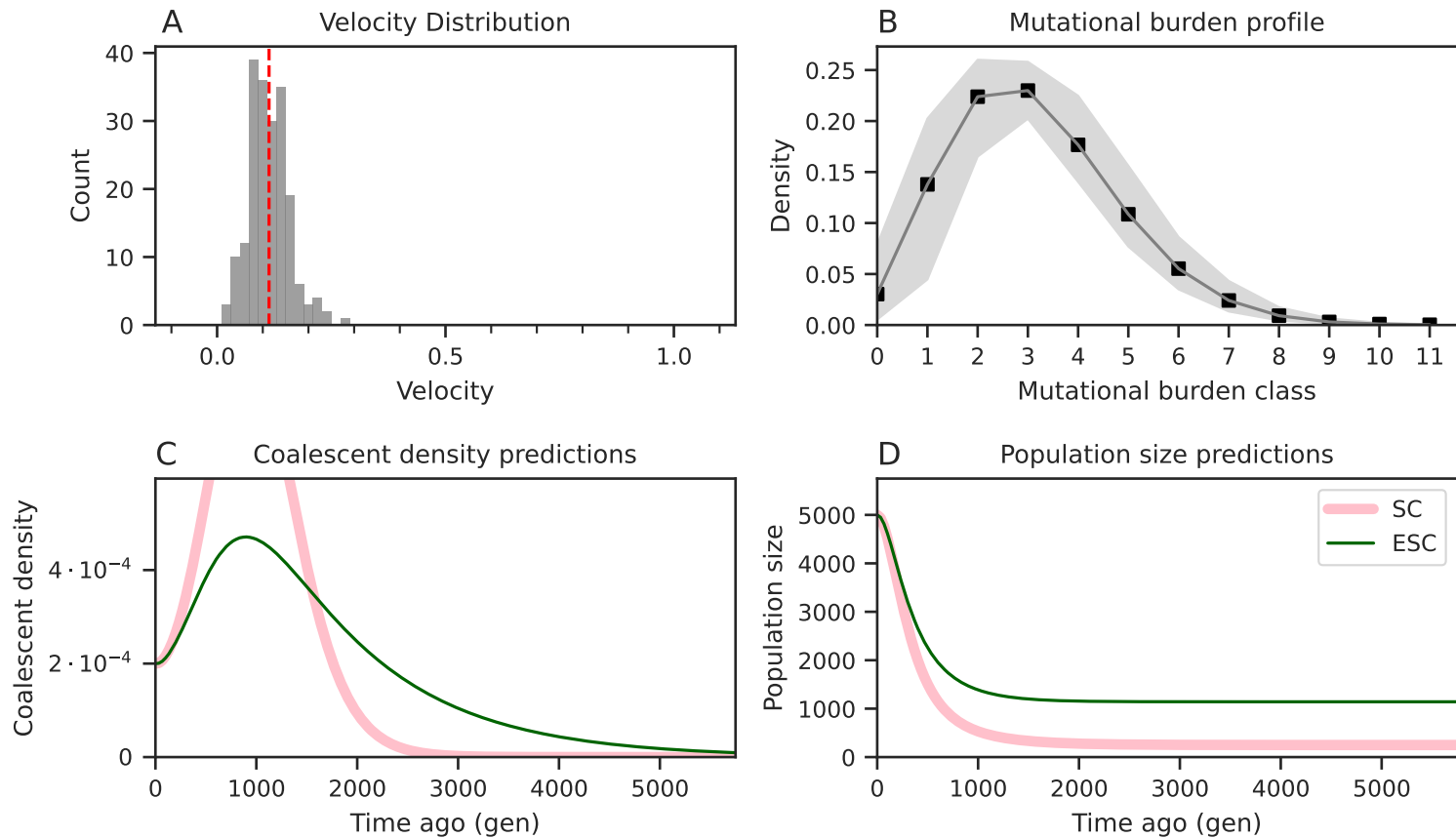
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity= 0.0152 ± 0.0083 , $n = 100$ simulations



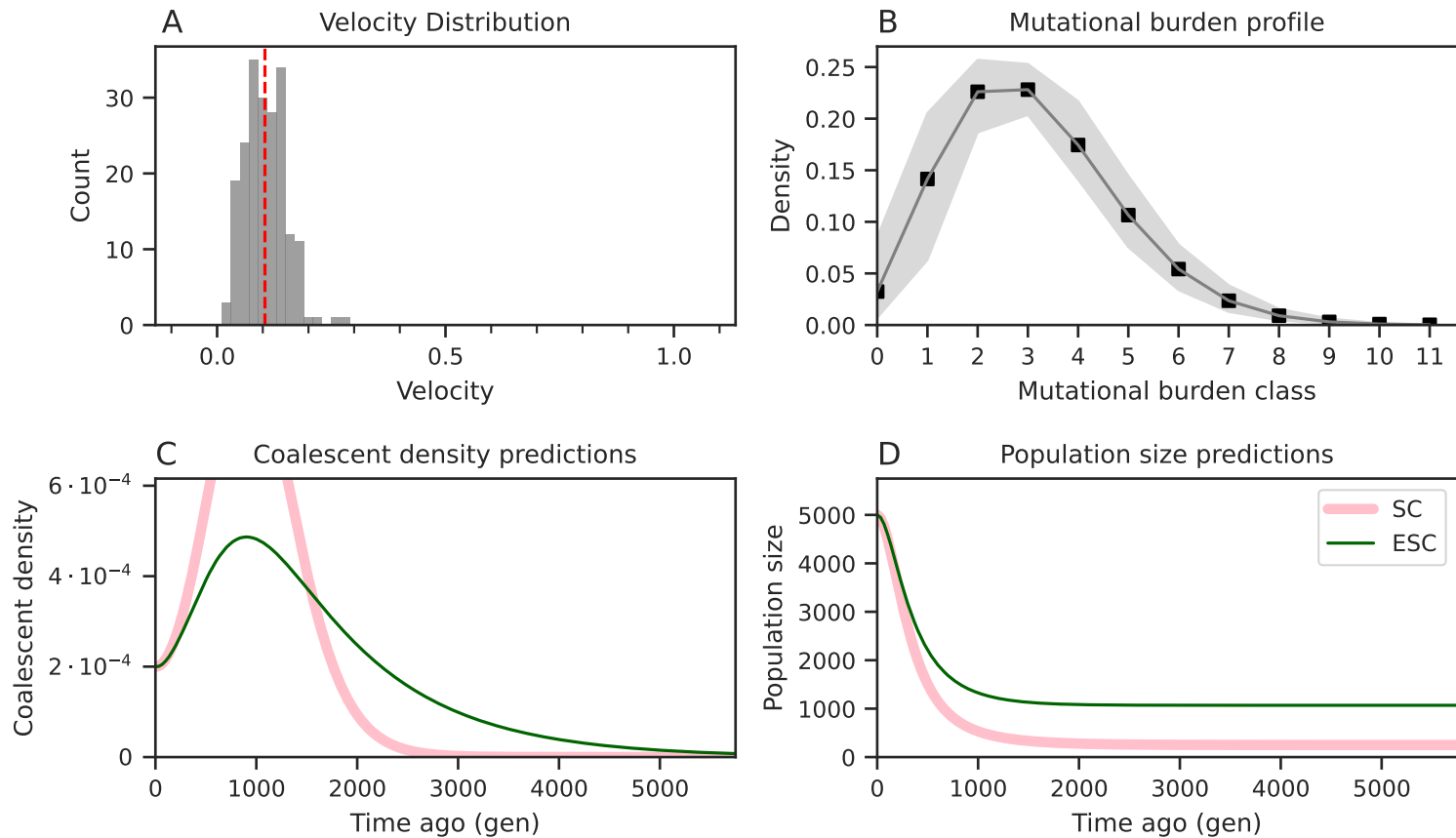
$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 0.498$
Mean Velocity= 0.1102 ± 0.0477 , $n = 200$ simulations



$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.1137 ± 0.0428 , $n = 200$ simulations

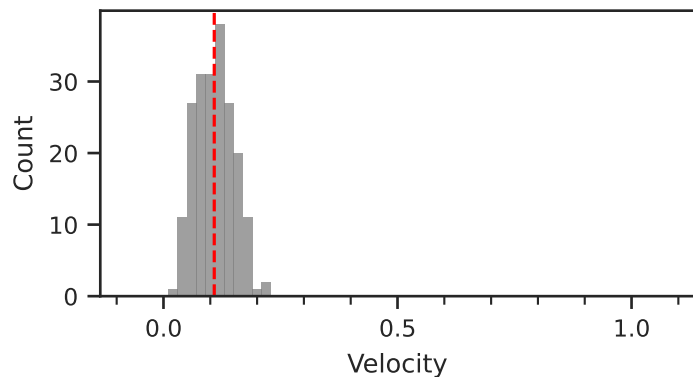


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity = 0.1047 ± 0.0437 , $n = 200$ simulations

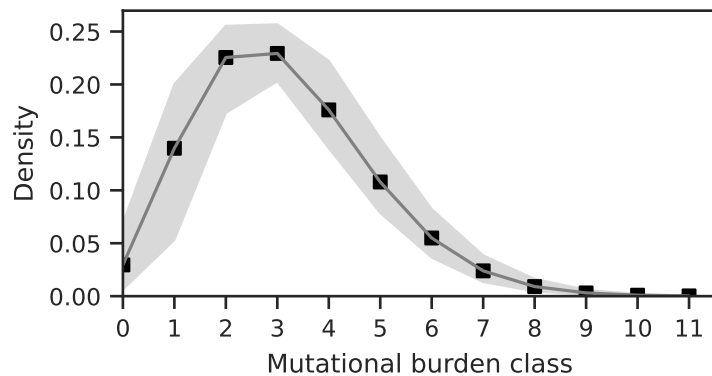


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity = 0.1085 ± 0.0407 , $n = 200$ simulations

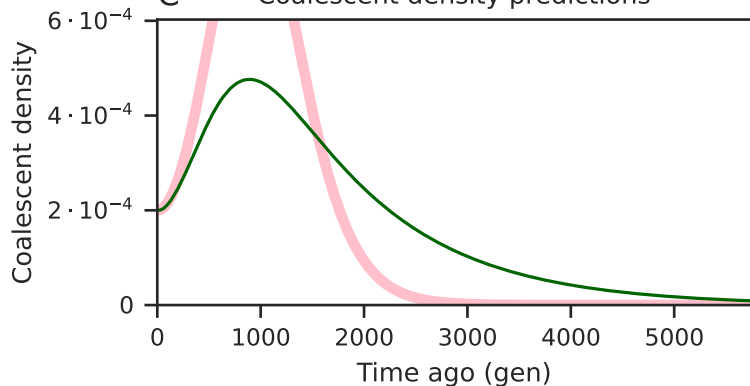
A Velocity Distribution



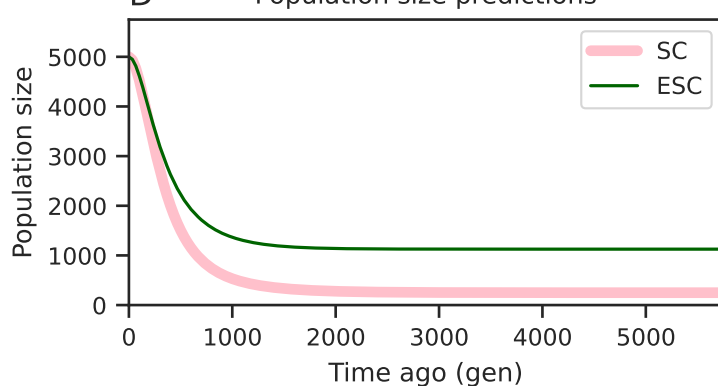
B Mutational burden profile



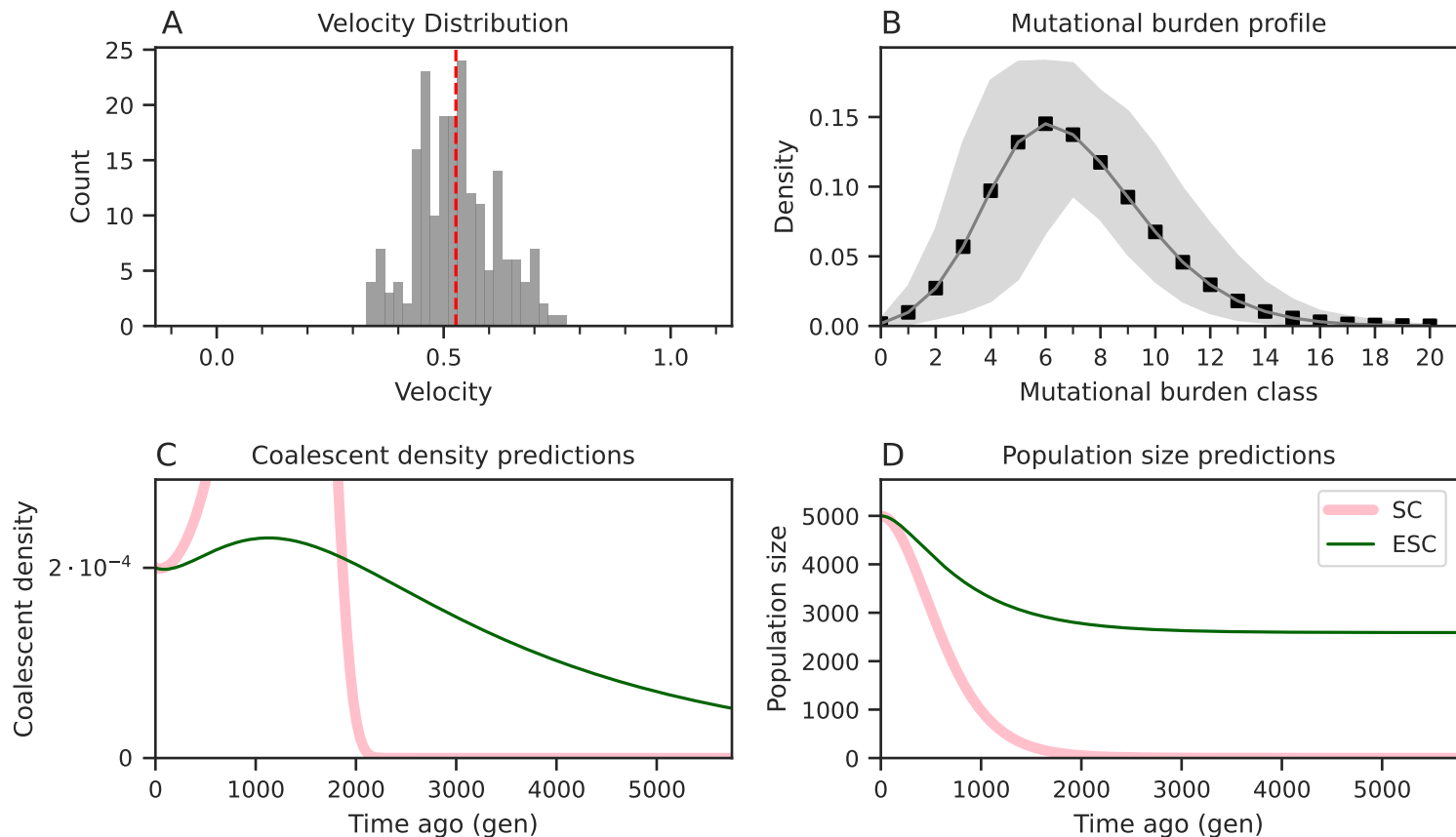
C Coalescent density predictions



D Population size predictions

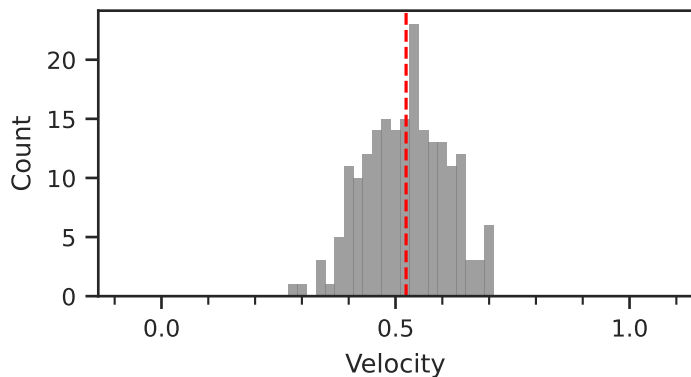


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5269 ± 0.0897 , $n = 200$ simulations

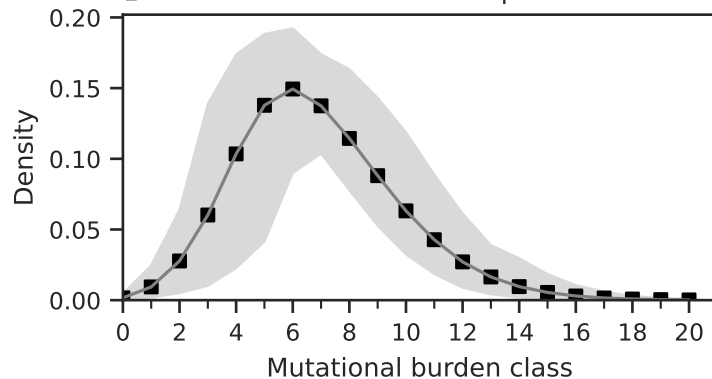


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5226 ± 0.0865 , $n = 200$ simulations

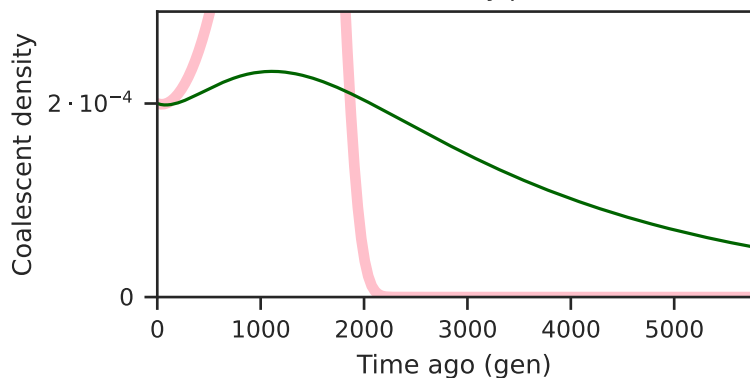
A Velocity Distribution



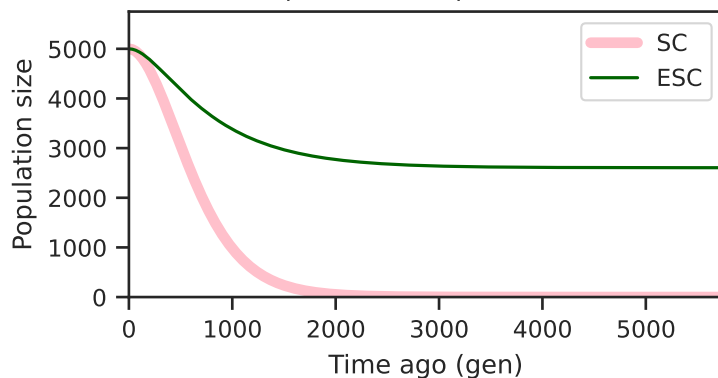
B Mutational burden profile



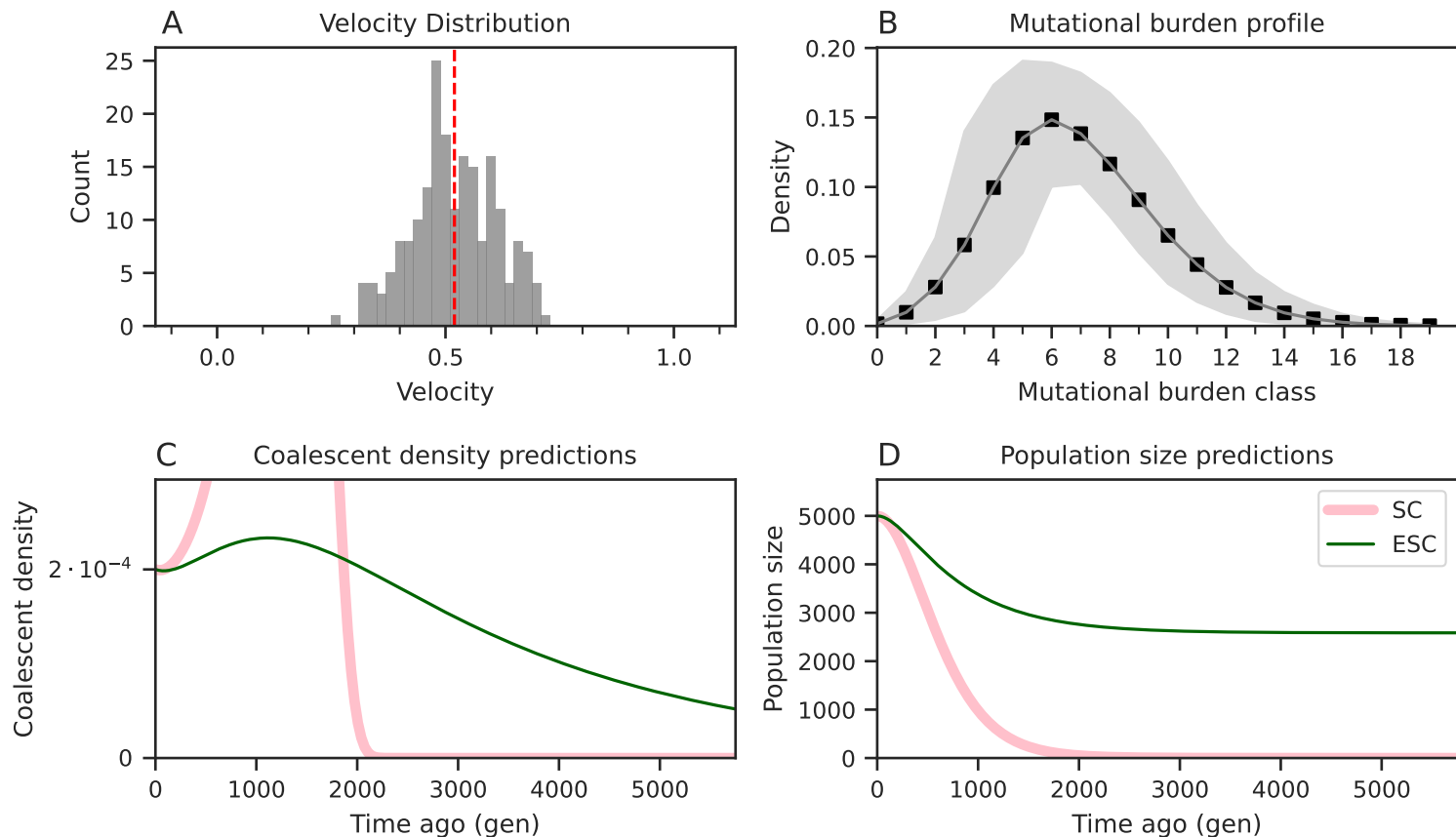
C Coalescent density predictions



D Population size predictions



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5196 ± 0.0921 , $n = 200$ simulations



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5148 ± 0.0881 , $n = 200$ simulations

